W.S.74. Pressure.

Name .....

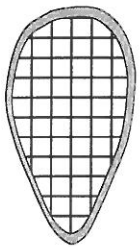
Pressure is the amount of force that is put onto a certain area.

$$\text{PRESSURE (N/m}^2\text{)} = \frac{\text{FORCE (N)}}{\text{AREA (m}^2\text{)}}$$

(Another unit for pressure is the pascal (Pa) — 1 N/m² = 1 Pa)

Small pressure.

A force is spread over a **large** area.



Snow shoes
These spread a person's weight over a large area which prevents them from sinking into soft snow.



Washers
A washer spreads the force from the nut which stops it being pulled into wood.

Large pressure.

A force is concentrated over a **small** area.



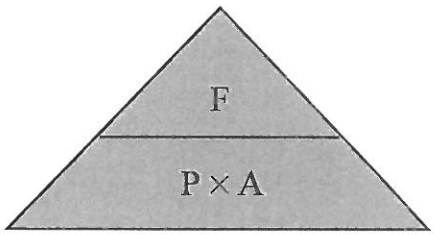
Knife edge – large pressure to cut into materials.



Stiletto heel
When all of the woman's weight is resting on the heel it produces a very large pressure. This can cause damage to floors with soft surfaces.

Working out force and area.

You can use the formula triangle on the right to work out pressure, force and area. For example, if you wish to work out force then place your finger over the force part (F) and you will see that force is pressure × area (P × A).



Exercise – Work out the answers to the questions below.

1) A man weighs 800N. The area of BOTH of his boots is 0.08m². What pressure does he place on the ground when he stands still?

$$\text{PRESSURE (N/m}^2\text{)} = \frac{\text{FORCE (N)}}{\text{AREA (m}^2\text{)}} = \frac{800\text{N}}{0.08\ \text{m}^2} = _____\ \text{N/m}^2$$

2) A woman weighs 500N. The area of ONE of her stiletto heels is 0.0002m². What pressure does she place on the ground when she puts her weight onto one heel?

$$\text{PRESSURE (N/m}^2\text{)} = \frac{\text{FORCE (N)}}{\text{AREA (m}^2\text{)}} = \frac{500\text{N}}{\quad} = _____\ \text{N/m}^2$$

3) The base of a suitcase has an area of 0.2m². It places a pressure of 700N/m² on the ground. What must the weight of the suitcase be?

force is pressure × area (P × A) = 700N/m² × _____ m² = _____ N

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